

Vraj Shaileshbhai Patel

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Computer Science M.S. candidate (May 2026) building and deploying **machine learning on large-scale clinical and financial data** across the full lifecycle — SQL feature engineering, gradient boosting and ensemble methods, rigorous leak-free evaluation (walk-forward, AUC-ROC, calibration), and AWS deployment. Strong in Python, SQL, and reproducible, version-controlled workflows; comfortable translating technical results for clinical and non-technical stakeholders.

EDUCATION

New Jersey Institute of Technology, Newark, NJ

Sep 2024 – May 2026

M.S. in Computer Science | GPA: 3.80 / 4.0

Gujarat Technological University (SAL Institute), Ahmedabad, India

Sep 2019 – May 2023

B.E. in Computer Engineering | GPA: 8.13 / 10

EXPERIENCE

New Jersey Institute of Technology — Graduate Researcher

Sep 2025 – Present

Clinical Data Analysis (PLCO Cancer-Screening Trial) | Advisor: Prof. Arashdeep Kaur

- Analyze the large-scale PLCO cancer-screening dataset (older-adult participants) in Python (Pandas), running EDA and statistical testing to rank candidate predictors and define the feature set used for downstream risk modeling.
- Process **confidential patient health and medical-imaging data** under strict privacy and governance standards, keeping sensitive records secure and analysis-ready without exposing protected health information.
- Develop preprocessing and feature-engineering pipelines and reduce redundant variables by 30% through data-driven feature selection.
- Evaluate outputs with AUC, F1, and Brier score and communicate findings to a research team through a standardized, reproducible workflow that both technical and non-technical members could act on.

PROJECTS

Lung Cancer Risk Prediction - Data Efficiency & Stability Analysis

GitHub

Python, scikit-learn, XGBoost, LightGBM, CatBoost, pandas, Matplotlib

- Designed a super-stacking ensemble combining six base learners (Logistic Regression, Random Forest, Gradient Boosting, XGBoost, LightGBM, CatBoost) with a logistic-regression meta-learner to predict lung cancer risk from 15 clinically interpretable variables in the PLCO Cancer Screening Trial.
- Ran a data-efficiency study training on progressively smaller fractions (100% → 10%) against a fixed, stratified 20% test set with 5 random seeds per fraction, reporting all metrics as mean ± std for reproducibility.
- Demonstrated strong robustness under data scarcity: AUC-ROC held near 0.836 and dropped only ~0.005 at 10% training data, with recall staying ~0.88.
- Mitigated severe class imbalance via `class_weight/scale_pos_weight` and tuned the decision threshold on validation only (F1-max), keeping the operating point stable near 0.30 while avoiding test-set leakage.

VolGuard - End-to-End Stock Volatility Forecasting

GitHub • Live Dashboard

Python, scikit-learn, XGBoost, SQL (SQLite), AWS (Lambda, S3), Terraform, Tableau

- Built an end-to-end ML pipeline forecasting 5-day realized volatility for five large-cap stocks: raw prices → SQL feature engineering → validated model → AWS Lambda → a live Tableau dashboard.
- Engineered features entirely in SQL (daily returns, 5/10/30-day rolling volatility, forward-5-day target with no look-ahead) and enforced strict chronological splits to prevent data leakage.
- Validated honestly with walk-forward testing across 4 out-of-sample folds, benchmarking Linear Regression, Random Forest, XGBoost, and GARCH(1,1) against a naive baseline; shipped the simplest model, cutting RMSE ~20% vs. baseline.
- Deployed to AWS with Terraform (infrastructure-as-code) — S3 + Lambda with API-key auth — serving predictions via CLI/SDK that feed the public dashboard.

Mini Amazon - Full-Stack E-Commerce

React, Tailwind CSS, FastAPI, MongoDB, JWT, WebSockets

- Constructed a scalable e-commerce app for 100+ users with product search, cart, and order tracking.
- Developed an admin dashboard supervising 500+ products with real-time order updates.

WORK EXPERIENCE

Mind Inventory, Ahmedabad, India

Jan 2023 – Apr 2023

Intern — Web Systems Design & Development

- Built and maintained scalable, database-driven web systems using PHP, MySQL (XAMPP), and front-end frameworks.
- Worked in an agile team to control relational databases and establish responsive UI across 3+ platforms, boosting page-load speed ~30%.
- Contributed to QA testing and bug-fix cycles, strengthening code reliability.

EliteEvince Technologies, Ahmedabad, India

Jun 2022 – Jul 2022

Web Developer Intern (PHP & MySQL)

- Translated real-world client requirements into deliverables under tight timelines.
- Resolved UI and documentation issues across 2 internal projects, increasing maintainability and usability.

SKILLS

Machine Learning

scikit-learn, XGBoost, Random Forest, Logistic Regression, cross-validation, hyperparameter tuning, ROC-AUC / model evaluation, feature selection

Data Analysis

Python (Pandas, NumPy), EDA, statistical analysis, Matplotlib

Healthcare Data

Clinical dataset analysis, confidential health-data handling, medical-imaging research

SQL & Databases

SQL, PL/SQL, MySQL, PostgreSQL, MongoDB

Tools

Git, GitHub, Docker, VS Code, Jupyter, Linux CLI, Excel

Languages

English, Hindi, Gujarati